

THRIFTY AI – CODING ASSESSMENT

Overview

This assignment evaluates your ability to design and implement a simplified Retrieval-Augmented Generation (RAG) system. You are expected to demonstrate strong fundamentals in Python, information retrieval, and applied machine learning concepts, along with clear thinking in system design.

Problem Statement

You are given a collection of documents and a set of user queries. Your task is to build a system that retrieves relevant information, constructs a structured prompt, generates an answer, and evaluates the quality of that answer.

Input Data

Documents (docs.json):

```
[
  {"id": 1, "text": "India launched Chandrayaan-3 in 2023."},
  {"id": 2, "text": "The mission successfully landed on the moon's south pole."},
  {"id": 3, "text": "ISRO is the Indian Space Research Organisation."},
  {"id": 4, "text": "Chandrayaan-2 had a partial failure during landing."}
]
```

Queries (queries.json):

```
[
  {"query": "Who launched Chandrayaan-3?"},
  {"query": "What happened in Chandrayaan-2?"},
  {"query": "Which organization is ISRO?"}
]
```

Required Tasks

1. Retrieval

```
def retrieve(query: str, docs: List[Dict], k: int = 2) -> List[str]:
```

Implement a retrieval mechanism to return the top-k most relevant documents for a given query. You may use TF-IDF or embedding-based similarity. The implementation must be in-memory.

2. Prompt Construction

```
def build_prompt(query: str, context: List[str]) -> str:
```

Construct a structured prompt that clearly separates context and question, ensuring that the answer generation step relies only on provided context.

3. Answer Generation

```
def generate_answer(prompt: str) -> str:
```

Simulate a language model by generating an answer using rule-based logic, extraction, or heuristics. The answer should be grounded in the provided context.

4. Evaluation

```
def evaluate(answer: str, context: List[str]) -> float:
```

Implement a scoring function that evaluates the answer on a scale from 0 to 1 based on relevance and groundedness.

Expected Output Format

Query: ...

Retrieved Context: [...]

Answer: ...

Score: 0.xx

Deliverables

1. Source code in Python
2. README file explaining your approach, design choices, and assumptions
3. Instructions to run the code
4. Sample outputs for the provided queries

Evaluation Criteria

- Retrieval quality and correctness
- Clarity and robustness of prompt design
- Correctness and grounding of generated answers
- Quality of evaluation methodology
- Code structure, readability, and documentation

Constraints

- The solution must be implemented in Python.

- External vector databases are not allowed.
- Use of external APIs is not permitted for answer generation.